

Brian Woods

Staff / Principal ML & Systems Engineer

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SUMMARY

Engineer with 30 years in production systems, now working at the intersection of ML research and systems engineering. I design experiments with pre-registered hypotheses and causal methods, build the reproducible pipelines that run them, and publish the results including the negative ones at brian.biz. My live Solana trading platform trains and serves reinforcement-learning and gradient-boosted models under drift monitoring, champion/challenger evaluation, and fail-closed gating, all built and operated solo; its trading P&L to date is flat to negative and reported publicly, because the measurement is the point. Four leadership stints (CTO, two Director of Engineering roles, head of growth engineering) hiring engineers and setting standards. I build with AI agents across the full engineering loop daily, using an open-source human-gated pipeline I wrote.

RESEARCH & SELECTED WORK

Block-Propagation Economics, two published studies · 2026 (brian.biz/optimum)

- Causal studies of propagation latency and validator revenue on Ethereum (216,000 mainnet slots; regression discontinuity at the 4-second attestation deadline) and Solana (within-leader-window fixed effects on data from my own collector node; +0.00372 SOL per 100 ms of earlier completion). Includes a 245-day deployment analysis reporting a negative result as such. Automated pipeline re-estimates and republishes daily; methods defended live in technical interviews.

Agent Benchmark Reliability Study · 2026 (brian.biz/deccan)

- Are agent benchmark scores stable enough to trust? 156 episodes across three procedurally generated, programmatically verified task families, perturbed along six axes that should not matter (paraphrase, tool renaming, distractors, resampling, temperature, order). Every axis shifted scores, and detecting a 3-point gap between models requires hundreds to thousands of instances: faithful comparisons cost one to two orders of magnitude more than they are typically given. Pre-registered hypotheses; full pipeline and tests published.

Audio Training-Data Value Study · 2026, in progress (brian.biz/david.ai)

- Pre-registered ablation of ASR training-data value: word error rate versus quantity, label noise, acoustic quality, and speaker diversity, across 35 planned Whisper-small fine-tuning runs and nine hypotheses. Experimental pipeline complete and verified by 213 tests; fine-tunes await GPU compute, and the page says exactly that.

Autonomous Trading Platform on Solana (sole owner-operator) · 2023-Present

- Real-time production ML and agents on real money: gRPC/Geyser streaming ingest at 15M+ transactions per run, multi-region execution with sub-second confirmation, Rust hot-path extension modules in the Python interpreter, RL training (PPO, CQL, behavioral cloning) and gradient-boosted models with staged shadow rollout, fail-closed gating, and a public live operations dashboard (brian.biz/leeroy).

Agentic Workflow, open source · (github.com/brianosaurus/agentic-workflow)

- A human-gated build pipeline for coding agents: one agent plans, implements, and verifies; a second reviews adversarially in a read-only sandbox; a human approves at every gate, with SHA-256-locked approvals and speculative execution of the next stage during review.

VitalFoods.ai, ML Recommendation Platform (UC Berkeley capstone, team project; live)

- Designed the original RotatE knowledge-graph embedding models and built the full production website; a teammate led embedding training. Ran all SageMaker and EC2 infrastructure.

EXPERIENCE

Independent Engineer & Data Scientist, Trading Systems & Published Research (self-employed) · 2023-Present

The systems and studies above, built concurrently with the UC Berkeley M.S. Started with one collaborator who has since moved on.

Chief Technology Officer, AGEN.AI (bootstrapped, never raised funding) · 2023-2024

Led architecture and delivery of the company's AI product across a small team; set engineering standards and hired.

Senior Software Developer, Umee (company wound down) · 2022

Built core infrastructure for a blockchain staking and lending platform in Go: collateralization, borrow limits, interest accrual.

Senior Protocol Engineer, Sifchain (company wound down) · 2021-2022

Hardened inter-blockchain communication (IBC) in production. Go and Python.

Core Developer, Augur Project · 2017-2020

Built decentralized prediction-market infrastructure (TypeScript/JavaScript).

Core Backend Developer, Dispatch Labs · 2015-2017

Owned the EVM implementation and its correctness.

Director of Engineering, HelloBit · 2013-2015

Owned architecture, hiring, and delivery for a Bitcoin remittance startup moving real money across borders.

Director of Engineering, AnythingSocial · 2010-2013

Led delivery of mobile and Rails consumer applications across distributed teams.

Head of Growth & International, Bebo, Inc. · 2005-2009

Ran growth engineering at one of the world's largest social networks at its peak (~250K registrations/day).

Software Engineer, Tickle · 2001-2005

Built a content-discovery product driven by user behavior data.

Senior Engineer, Yahoo Mail, Yahoo Inc. · 1996-2001

Wrote the C++ mail transport agent handling 6B+ messages/month: queueing, backpressure, and delivery guarantees at planetary scale, plus anti-spam classification (6 patents granted).

EDUCATION

M.S., Information & Data Science (AI concentration), UC Berkeley · 2024-2026

Capstone shipped as VitalFoods.ai (above). Completed while building and operating the trading platform.

B.S., Computer Science & Economics, University of Colorado Boulder · 1996**PATENTS & CERTIFICATIONS**

- 6 U.S. patents in bulk-email processing and anti-spam (Yahoo Mail).
- Cryptography I (Stanford) · GCP Big Data & ML Fundamentals.

SKILLS

ML research & evaluation: experiment design and pre-registration, causal inference (RDD, difference-in-differences, fixed-effects panels, placebo tests), benchmark and eval construction, verifiable-reward task design, bootstrap uncertainty and power analysis, honest reporting of negative results

ML engineering: PyTorch, Hugging Face transformers/datasets, RL training (PPO, CQL, behavioral cloning), LightGBM/XGBoost, PyMC, feature pipelines, drift monitoring, champion/challenger, AWS SageMaker and EC2 run hands-on, production MLOps built from scratch

Systems: real-time streaming (gRPC), multi-region execution, queueing and backpressure, fail-closed design, observability and public telemetry, resumable manifest-driven pipelines; Python, Rust (extension modules in the Python interpreter), Go, C++, SQL, TypeScript/JavaScript

Leadership & practice: hired and mentored across four leadership stints; standards for research rigor, documentation, and reproducibility; AI agents in the full engineering loop daily (spec, adversarial plan review, build, test); MCP server development